

MH3100 Real Analysis I

Midterm Practice 2 (Solutions)

AY 2026/2027, Semester 1 – QRS original

Quantitative Research Society @NTU

September 29, 2026

TIME ALLOWED: 60 minutes.

Instructions to Students

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the box provided. Use the **facing page** for more space.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, **unless you are asked by the question to prove the result.**

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24
Awarded Marks					

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Question 1**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If $\{a_n\}$ is a bounded sequence such that every convergent subsequence of $\{a_n\}$ converges to the same real number L , then $\{a_n\}$ converges to L .
- (b) (2 marks) There is a monotone sequence that has a convergent subsequence but is itself divergent.
- (c) (2 marks) If $\{a_n\}$ and $\{b_n\}$ are Cauchy sequences with $b_n \neq 0$ for every $n \in \mathbb{N}$, then $\left\{\frac{a_n}{b_n}\right\}$ is a Cauchy sequence.

Solution

- (a) True

Justification. Suppose, for the sake of contradiction, that $\{a_n\}$ does not converge to L . Then, by the definition of the limit, there is a number $\varepsilon > 0$ such that infinitely many terms of $\{a_n\}$ satisfy $|a_n - L| \geq \varepsilon$. Let $\{a_{n_k}\}$ denote the subsequence consisting of exactly these terms; it is bounded, since $\{a_n\}$ is bounded. For each $k \in \mathbb{N}$, either $a_{n_k} - L \geq \varepsilon$ or $a_{n_k} - L \leq -\varepsilon$; since there are infinitely many k in total, at least one of these two cases holds for infinitely many k . Pass to that further subsequence, and call it $\{a_{n_{k_j}}\}$; it is still bounded. By the Bolzano-Weierstrass Theorem, $\{a_{n_{k_j}}\}$ has a convergent subsequence $\{a_{n_{k_{j_i}}}\}$, and this is also a subsequence of the original sequence $\{a_n\}$. Let L' denote its limit. Since every term of $\{a_{n_{k_{j_i}}}\}$ eventually satisfies $a_{n_{k_{j_i}}} - L \geq \varepsilon$ (or, in the other case, $L - a_{n_{k_{j_i}}} \geq \varepsilon$), the Order Limit Theorem implies $L' - L \geq \varepsilon$ (respectively, $L - L' \geq \varepsilon$); either way, $L' \neq L$. Therefore we have exhibited a convergent subsequence of $\{a_n\}$ whose limit is not L , contradicting the hypothesis. Hence $\{a_n\}$ converges to L . $\square$

- (b) False

Justification. The Monotone Convergence Lemma states that if a sequence is eventually monotone and has a convergent subsequence, then the sequence itself is convergent. Therefore no monotone sequence can have a convergent subsequence while itself being divergent, so the statement is false. $\square$

- (c) False

Justification. Take $a_n := 1$ for every $n \in \mathbb{N}$ and $b_n := \frac{1}{n}$ for every $n \in \mathbb{N}$. The constant sequence $\{a_n\}$ is convergent (to 1), hence Cauchy by the Cauchy Criterion. The sequence $\{b_n\}$ converges to 0, hence is also Cauchy by the Cauchy Criterion, and $b_n \neq 0$ for every $n \in \mathbb{N}$. However, $\frac{a_n}{b_n} = n$ for every $n \in \mathbb{N}$, and $\{n\}$ is unbounded. Since every convergent sequence is bounded (Boundedness of Convergent Sequences), the contrapositive shows that $\{n\}$ is divergent; by the Cauchy Criterion, a divergent sequence cannot be Cauchy. Hence $\left\{\frac{a_n}{b_n}\right\}$ is not Cauchy, so the statement is false. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- (a) **(2 marks)** 2 marks for ☐ True; 0 marks for ☐ False. No justification is required on the actual exam.
- (b) **(2 marks)** 2 marks for ☐ False; 0 marks for ☐ True.
- (c) **(2 marks)** 2 marks for ☐ False; 0 marks for ☐ True.

Typical deductions (if justification is examined, e.g. in a tutorial recap): in (a), asserting the result without reference to a compactness-type tool (Bolzano-Weierstrass); in (c), attempting a "proof" instead of noticing a quotient can be unbounded; confusing Cauchy with merely bounded.

Question 2

(6 Marks)

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges.
 (b) (2 marks) Every finite subset of $\mathbb{R}$ is a closed set.
 (c) (2 marks) $\overline{\mathbb{Q} \cap [0, 1]} = [0, 1]$.

Solution

- (a) False

Justification. (*Scratch work – not part of the written answer.*) Try an alternating series whose terms are not absolutely summable, e.g. $a_n := \frac{(-1)^n}{\sqrt{n}}$; then $a_n^2 = \frac{1}{n}$ is the harmonic series, which diverges.

Let $a_n := \frac{(-1)^n}{\sqrt{n}}$ for each $n \in \mathbb{N}$. The sequence $\left\{\frac{1}{\sqrt{n}}\right\}$ is decreasing and converges to 0, so the Alternating Series Test implies that $\sum_{n=1}^{\infty} a_n$ converges. However, $a_n^2 = \frac{1}{n}$ for every $n \in \mathbb{N}$, so $\sum_{n=1}^{\infty} a_n^2$ is the harmonic series $\sum_{n=1}^{\infty} 1/n$. For each $m \in \mathbb{N}$, the partial sums satisfy

$$s_{2m} - s_m = \frac{1}{m+1} + \frac{1}{m+2} + \cdots + \frac{1}{2m} \geq m \cdot \frac{1}{2m} = \frac{1}{2},$$

since each of the m terms is at least $\frac{1}{2m}$. Taking $\varepsilon = \frac{1}{2}$, this shows that the sequence of partial sums of $\sum_{n=1}^{\infty} 1/n$ does not satisfy the Cauchy Criterion for Series, so $\sum_{n=1}^{\infty} 1/n$ diverges. Hence $\sum_{n=1}^{\infty} a_n$ converges while $\sum_{n=1}^{\infty} a_n^2$ diverges, so the statement is false. $\square$

- (b) True

Justification. Let $F = \{x_1, x_2, \dots, x_m\} \subseteq \mathbb{R}$ be a finite set (the case $F = \emptyset$ is vacuous, since $\emptyset$ has no limit points). Let $a \in \mathbb{R}$ be arbitrary; we show a is not a limit point of F . If $F \setminus \{a\}$ is empty, then there is no $x \in F \setminus \{a\}$ at all, so a cannot satisfy the definition of a limit point of F . If $F \setminus \{a\}$ is nonempty, it is a finite set of reals other than a , so $d := \min_{x \in F \setminus \{a\}} |x - a|$ exists and $d > 0$. Taking $\varepsilon = d$ in the definition of a limit point, there is no $x \in F \setminus \{a\}$ with $|x - a| < d$, since every such x satisfies $|x - a| \geq d$ by the choice of d . Hence a is not a limit point of F either way. Since $a \in \mathbb{R}$ was arbitrary, F has no limit points at all; therefore F vacuously contains all its limit points, so F is closed. $\square$

- (c) True

Justification. (*Scratch work – not part of the written answer.*) Show $[0, 1]$ is closed (via the Order Limit Theorem) so the closure cannot exceed $[0, 1]$; then exhibit, for each $x \in [0, 1]$, an explicit rational sequence $q_n := k_n/n$ (with $k_n := \max\{k \in \mathbb{N}_0 : k \leq nx\}$) in $[0, 1]$ converging to x .

We first show $[0, 1]$ is closed. Let a be a limit point of $[0, 1]$; by the Sequential Criterion for Limit Points, there is a sequence $\{x_n\}$ contained in $[0, 1] \setminus \{a\}$ with $a = \lim x_n$. Since $x_n \geq 0$ eventually (indeed always) and $x_n \leq 1$ eventually, the Order Limit Theorem gives $a \geq 0$ and $a \leq 1$, so $a \in [0, 1]$. Hence every limit point of $[0, 1]$ lies in $[0, 1]$, i.e. $[0, 1]$ is closed.

($\subseteq$) Let $a \in \overline{\mathbb{Q} \cap [0, 1]}$. By the ε -Criterion for Closures, for every $\varepsilon > 0$ there is some $x \in \mathbb{Q} \cap [0, 1]$ with $|x - a| < \varepsilon$; since $x \in [0, 1]$ also, a satisfies the ε -Criterion for the closure of $[0, 1]$, so $a \in \overline{[0, 1]}$. Since $[0, 1]$ is closed, the Closure Criterion for Closed Sets gives $\overline{[0, 1]} = [0, 1]$, so $a \in [0, 1]$. Hence $\overline{\mathbb{Q} \cap [0, 1]} \subseteq [0, 1]$.

($\supseteq$) Let $x \in [0, 1]$ be arbitrary. For each $n \in \mathbb{N}$, let $S_n := \{k \in \mathbb{N}_0 : k \leq nx\}$. Since $x \geq 0$, we have $0 \in S_n$, so S_n is nonempty; and since $x \leq 1$, every $k \in S_n$ satisfies $k \leq nx \leq n$, so S_n is a subset of the finite set $\{0, 1, \dots, n\}$. Therefore S_n has a greatest element $k_n := \max S_n$, and $0 \leq k_n \leq n$. Define $q_n := k_n/n \in \mathbb{Q}$; then $0 \leq q_n \leq 1$, i.e. $q_n \in \mathbb{Q} \cap [0, 1]$. Since $k_n \in S_n$, we have $k_n \leq nx$; and since $k_n + 1 \notin S_n$, we have $k_n + 1 > nx$. Hence $0 \leq nx - k_n < 1$, so

$$|q_n - x| = \frac{|nx - k_n|}{n} < \frac{1}{n}.$$

Let $\varepsilon > 0$ be given. By the Archimedean Property, there is $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$. Then for every $n \geq N$, $|q_n - x| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. Hence $x = \lim q_n$, so, by the Sequential Criterion for Closures, $x \in \overline{\mathbb{Q} \cap [0, 1]}$. Since $x \in [0, 1]$ was arbitrary, $[0, 1] \subseteq \overline{\mathbb{Q} \cap [0, 1]}$.

Combining both inclusions, $\overline{\mathbb{Q} \cap [0, 1]} = [0, 1]$. □

Marking scheme (6 marks)

QRS suggested marking scheme.

- (a) **(2 marks)** 2 marks for False; 0 marks for True.
- (b) **(2 marks)** 2 marks for True; 0 marks for False.
- (c) **(2 marks)** 2 marks for True; 0 marks for False.

Typical deductions (if justification is examined): in (a), assuming convergence of $\sum_{n=1}^{\infty} a_n$ is absolute; in (b), confusing "closed" with "bounded" or "compact"; in (c), forgetting to prove the reverse inclusion (that the closure cannot exceed $[0, 1]$).

Question 3

(6 Marks)

In this question, A is a nonempty set of real numbers that is bounded below, and $c > 0$. Write down the definition of $s = \inf A$, and use the definition to show that

$$\inf(cA) = c \inf A, \quad \text{where } cA := \{ca : a \in A\}.$$

Do not use the ε -Criterion for Suprema and Infima.

Solution

Method 1: Direct verification of the two defining properties

Idea. (*Scratch work – not part of the written answer.*) Since $c > 0$, multiplying an inequality by c preserves its direction, so $a \geq s$ for all $a \in A$ gives $ca \geq cs$ for all $ca \in cA$; conversely, dividing a lower bound of cA by c turns it into a lower bound of A , which cannot exceed s .

Proof. A number $s \in \mathbb{R}$ is said to be the infimum (also known as the greatest lower bound) of a set $A \subseteq \mathbb{R}$, written $s = \inf A$, if (1) s is a lower bound for A , that is, $a \geq s$ for every $a \in A$; and (2) s is not less than any lower bound for A , that is, for every lower bound L of A , we have $s \geq L$.

Let $s = \inf A$. We show that cs satisfies both defining properties of $\inf(cA)$.

Step 1: cs is a lower bound for cA . Let $a \in A$ be arbitrary. Since s is a lower bound for A , we have $a \geq s$. Since $c > 0$, multiplying both sides of this inequality by c preserves its direction, so $ca \geq cs$. Every element of cA is of the form ca for some $a \in A$; therefore cs is a lower bound for cA .

Step 2: cs is not less than any lower bound of cA . Let L be an arbitrary lower bound for cA . Let $a \in A$ be arbitrary; then $ca \in cA$, so, since L is a lower bound for cA , we have $ca \geq L$. Since $c > 0$, dividing both sides of this inequality by c preserves its direction, so $a \geq L/c$. Since $a \in A$ was arbitrary, L/c is a lower bound for A . Since $s = \inf A$ is not less than any lower bound of A , we have $s \geq L/c$. Multiplying both sides of this inequality by $c > 0$ (which again preserves its direction) gives $cs \geq L$. Since L was an arbitrary lower bound for cA , cs is not less than any lower bound of cA .

By Steps 1 and 2, cs satisfies both defining properties of the infimum of cA . Therefore $\inf(cA) = cs = c \inf A$. $\square$

Common slip. Proving only Step 1 (that cs is a lower bound) and stopping there, without also showing that cs is the *greatest* lower bound (Step 2), leaves the proof incomplete since any smaller number is also a lower bound. Another common slip is dividing or multiplying an inequality by c without noting that $c > 0$ is what preserves (rather than reverses) the direction of the inequality; this matters because the identical argument fails for $c < 0$.

Marking scheme (6 marks)

QRS suggested marking scheme.

- **2 marks:** definition of $s = \inf A$ stated correctly, with both clauses (lower bound; not less than any lower bound) and no quantifier-order errors.
- **3 marks:** complete proof of both Step 1 (1 mark: cs is a lower bound for cA) and Step 2 (2 marks: cs is the greatest lower bound of cA , including the division step and the correct use of $s = \inf A$).

- **1 mark:** the three writing expectations are met throughout (complete sentences; related statements joined by connectives; inference marked with "therefore/hence/so").

Typical deductions: omitting Step 2 entirely; using the ε -Criterion for Suprema and Infima despite the prohibition; silently assuming $c > 0$ preserves inequality direction without stating it; sign errors when dividing by c .

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Question 4

(6 Marks)

In this question, $\{a_n\}$ is a sequence of nonnegative real numbers. Write down the definition of $a = \lim a_n$, and use the definition to show that if $a = \lim a_n$, then $\sqrt{a} = \lim \sqrt{a_n}$.

Do not use the Algebraic Limit Theorem.

Solution

Method 1: Rationalisation, split into the cases $a = 0$ and $a > 0$

Idea. (*Scratch work – not part of the written answer.*) For $a > 0$, rationalise: $\sqrt{a_n} - \sqrt{a} = (a_n - a)/(\sqrt{a_n} + \sqrt{a})$, and since $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a} > 0$, this gives $|\sqrt{a_n} - \sqrt{a}| \leq |a_n - a|/\sqrt{a}$, which can be made small by choosing $|a_n - a| < \varepsilon\sqrt{a}$. This breaks down when $a = 0$ (division by $\sqrt{a} = 0$), so that case is handled directly: $a_n < \varepsilon^2 \implies \sqrt{a_n} < \varepsilon$.

Definition. A sequence $\{a_n\}$ is said to have the limit $a \in \mathbb{R}$, written $a = \lim a_n$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that, for all $n \in \mathbb{N}$ with $n \geq N$, we have $|a_n - a| < \varepsilon$.

Proof. Since $a_n \geq 0$ for every $n \in \mathbb{N}$, the sequence $\{a_n\}$ eventually satisfies $a_n \geq 0$; therefore, by the Order Limit Theorem, $a = \lim a_n \geq 0$. So $\sqrt{a}$ is a well-defined real number, and we consider two cases.

Case $a = 0$. Let $\varepsilon > 0$ be given. Since $a = 0 = \lim a_n$, the definition of the limit gives $N \in \mathbb{N}$ such that, for all $n \in \mathbb{N}$ with $n \geq N$, we have $|a_n - 0| < \varepsilon^2$, that is, $0 \leq a_n < \varepsilon^2$, using $a_n \geq 0$. Since the square root of a nonnegative real number is an increasing function of that number, $0 \leq a_n < \varepsilon^2$ implies $0 \leq \sqrt{a_n} < \varepsilon$, that is, $|\sqrt{a_n} - 0| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this shows $0 = \lim \sqrt{a_n}$, so $\sqrt{a} = \lim \sqrt{a_n}$ in this case.

Case $a > 0$. Then $\sqrt{a} > 0$. For every $n \in \mathbb{N}$, since $a_n \geq 0$, we have $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a} > 0$, so

$$\sqrt{a_n} - \sqrt{a} = \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} = \frac{a_n - a}{\sqrt{a_n} + \sqrt{a}},$$

and therefore, since $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a} > 0$,

$$|\sqrt{a_n} - \sqrt{a}| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \leq \frac{|a_n - a|}{\sqrt{a}}.$$

Let $\varepsilon > 0$ be given. Since $\varepsilon\sqrt{a} > 0$ and $a = \lim a_n$, the definition of the limit (applied with $\varepsilon\sqrt{a}$ in place of ε) gives $N \in \mathbb{N}$ such that, for all $n \in \mathbb{N}$ with $n \geq N$, we have $|a_n - a| < \varepsilon\sqrt{a}$. Therefore, for all $n \geq N$,

$$|\sqrt{a_n} - \sqrt{a}| \leq \frac{|a_n - a|}{\sqrt{a}} < \frac{\varepsilon\sqrt{a}}{\sqrt{a}} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this shows $\sqrt{a} = \lim \sqrt{a_n}$ in this case as well.

In both cases, $\sqrt{a} = \lim \sqrt{a_n}$. □

Common slip. Dividing by $\sqrt{a}$ without first checking $a > 0$ (this step is undefined, or divides by 0, when $a = 0$); forgetting to justify $a \geq 0$ before writing $\sqrt{a}$ at all (the Order Limit Theorem step); and choosing N that is allowed to depend on n rather than only on ε .

Marking scheme (6 marks)

QRS suggested marking scheme.

- **2 marks:** definition of $a = \lim a_n$ stated correctly, with the quantifiers in the correct order (N may depend on ε , not vice versa).
- **1 mark:** correctly establishes $a \geq 0$ via the Order Limit Theorem before defining $\sqrt{a}$.
- **2 marks:** correct treatment of both cases ($a = 0$ and $a > 0$), including the rationalisation identity and a valid choice of N for the given ε .
- **1 mark:** the three writing expectations are met throughout (complete sentences; related statements joined by connectives; inference marked with "therefore/hence/so").

Typical deductions: omitting the case $a = 0$; using the Algebraic Limit Theorem or citing "continuity of $\sqrt{\cdot}$ " despite the prohibition; dividing by $\sqrt{a}$ without justifying $a > 0$ first; choosing N depending on n .